

SMART AUTOMATED MEDICINE DISPENSING AND ALERT SYSTEM FOR BLIND AND OLD AGE PEOPLE

EEB1323 – MICROPROCESSOR AND MICROCONTROLLER

BATCH NO : 12

END SEMESTER REVIEW

PRESENTED BY:

1. POOBALAN C [927623BEE069]
2. YUVARAJ M [927623BEE126]
3. SAKTHIVEL A [927623BEE306]

GUIDED BY:

Mr. CHOCKALINGAM. AL, M.TECH
ASSOCIATE PROFESSOR
DEPARTMENT OF EEE

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ABSTRACT



- THIS PROJECT PRESENTS A SMART AUTOMATED MEDICINE DISPENSING AND ALERT SYSTEM DESIGNED TO ASSIST ELDERLY AND VISUALLY IMPAIRED INDIVIDUALS IN MANAGING THEIR DAILY MEDICATION EFFECTIVELY. THE SYSTEM USES A CYLINDRICAL STORAGE MECHANISM TO ORGANIZE TABLETS FOR DIFFERENT TIME SCHEDULES SUCH AS MORNING, AFTERNOON, AND NIGHT. IT IS EQUIPPED WITH A MICROCONTROLLER, SERVO MOTOR, SENSORS, BUZZER, AND IOT MODULE TO AUTOMATE THE DISPENSING PROCESS.
- THE DEVICE OPERATES BY RELEASING THE CORRECT TABLET AT THE SCHEDULED TIME USING A SERVO MOTOR, WHILE A BUZZER ALERT NOTIFIES THE USER TO TAKE THE MEDICINE. SENSORS ARE USED TO DETECT WHETHER THE MEDICINE HAS BEEN TAKEN, ENSURING PROPER MONITORING. THE SYSTEM ALSO SENDS REAL-TIME NOTIFICATIONS TO CAREGIVERS THROUGH A MOBILE OR WEB APPLICATION, PROVIDING UPDATES ON MEDICATION STATUS.

INTRODUCTION



- Medication management is a major challenge for elderly and visually impaired individuals due to memory issues and difficulty in following schedules.
- Traditional methods like pill boxes, alarms, and manual reminders are unreliable and depend on human memory or caregiver support.
- Missing doses or taking medicines at the wrong time can lead to serious health problems and reduced treatment effectiveness.
- The advancement of IoT and automation provides a smart solution for accurate and timely medication management.
- The system enables automatic medicine dispensing at scheduled times such as morning, afternoon, and night.
- It includes components like a microcontroller, servo motor, cylindrical storage, sensors, and buzzer for efficient operation.

LITERATURE REVIEW

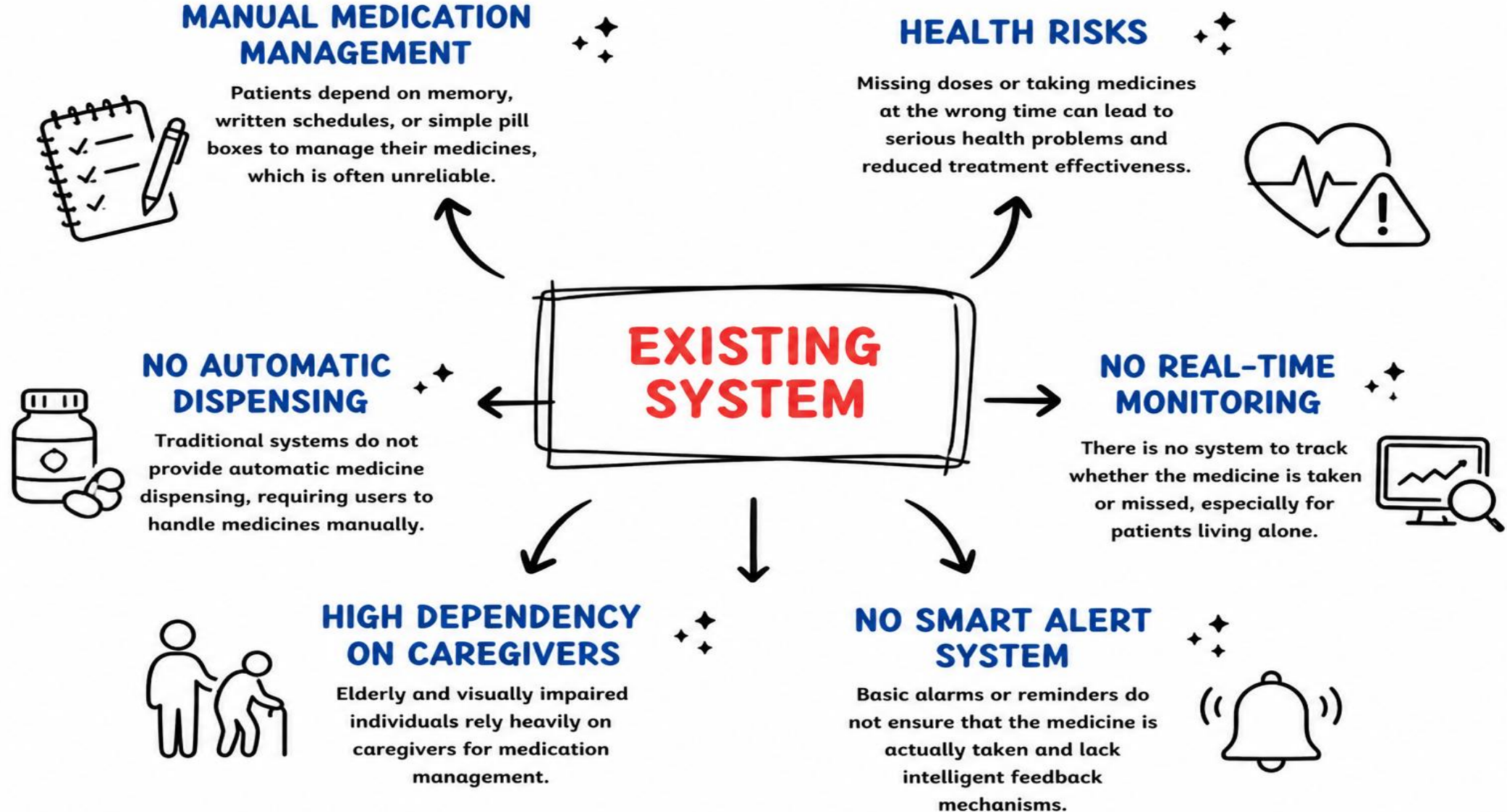
S. No	Title of Paper	Author(s)	Year of Publication	Description	Remarks
1	Smart Automated Medicine Reminder System	G. K. Patil, S. S. Patil	2019	IoT-based system that provides medicine reminders and health monitoring using sensors and alerts.	Focuses on reminders only; no automatic dispensing mechanism.
2	Smart Pill Box Using Arduino and GSM	S. K. Sharma, R. Kumar	2018	Pill box system with GSM alerts to notify users about medication timings.	Lacks real-time monitoring and feedback on medicine intake.
3	IoT Based Smart Healthcare System for Elderly	P. R. Dinesh, S. P. Chandra	2018	Uses IoT to monitor patient health and send alerts for medication schedules.	Does not include automated medicine dispensing.

LITERATURE REVIEW

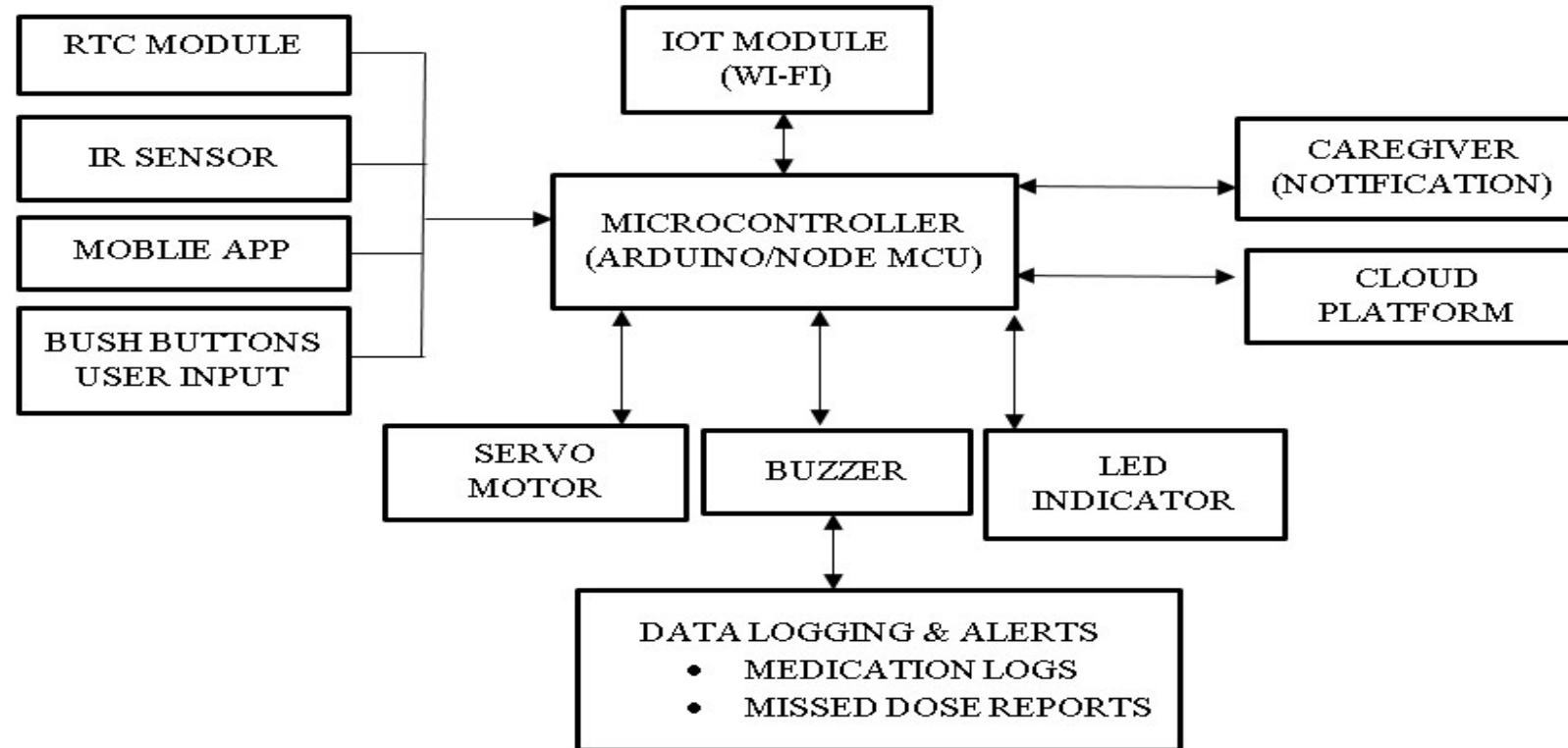


S. No	Title of Paper	Author(s)	Year of Publication	Description	Remarks
4	Sewage Cleaning and Management System	H. K. Patil, R. S. Yadav	2020	Provides alerts and notifications to caregivers when medicines are due or missed.	Limited automation; no intelligent dispensing system.
5	Automatic Pill Dispenser with SMS Alert	D. Malhotra, A. Kumar	2019	Microcontroller-based system that dispenses pills and sends SMS alerts.	Limited monitoring and no feedback mechanism for intake confirmation.

EXISTING SYSTEM



BLOCK DIAGRAM OF EXISTING SYSTEM

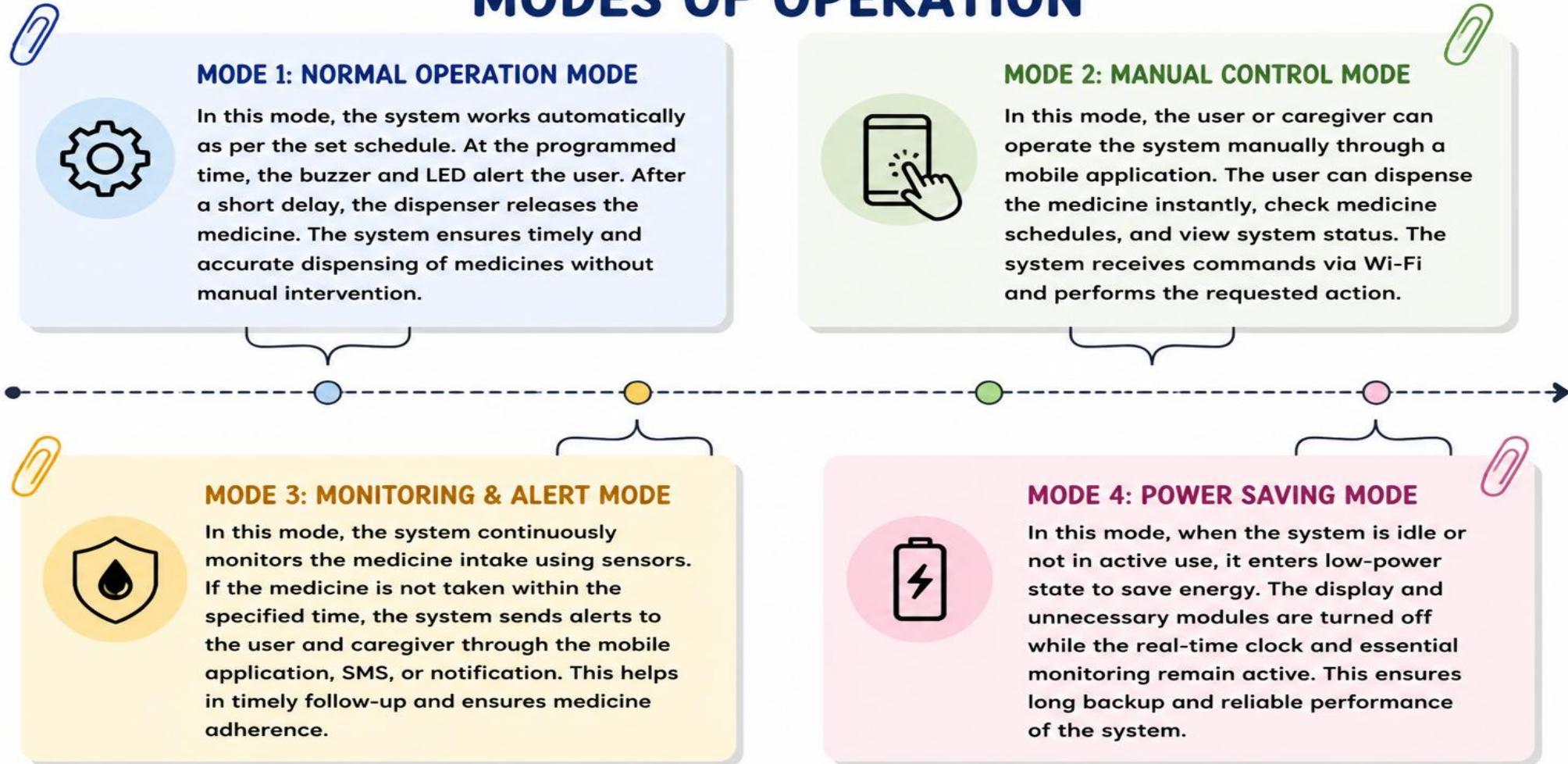


PROBLEM STATEMENT

- Medication management is a major challenge for elderly and visually impaired individuals, as they often forget to take medicines on time or take incorrect dosages. This can lead to serious health complications, reduced treatment effectiveness, and increased risk of medical emergencies.
- Existing methods such as pill boxes, alarms, and caregiver reminders are not reliable because they depend heavily on human memory and constant supervision. These approaches do not ensure whether the medicine is actually taken, making them ineffective for proper medication adherence.
- There is a need for an intelligent and automated system that can dispense medicines at the correct time, monitor whether the medicine is taken, and provide real-time notifications to caregivers, thereby improving patient safety and reducing dependency on manual support.

MODES OF OPERATION

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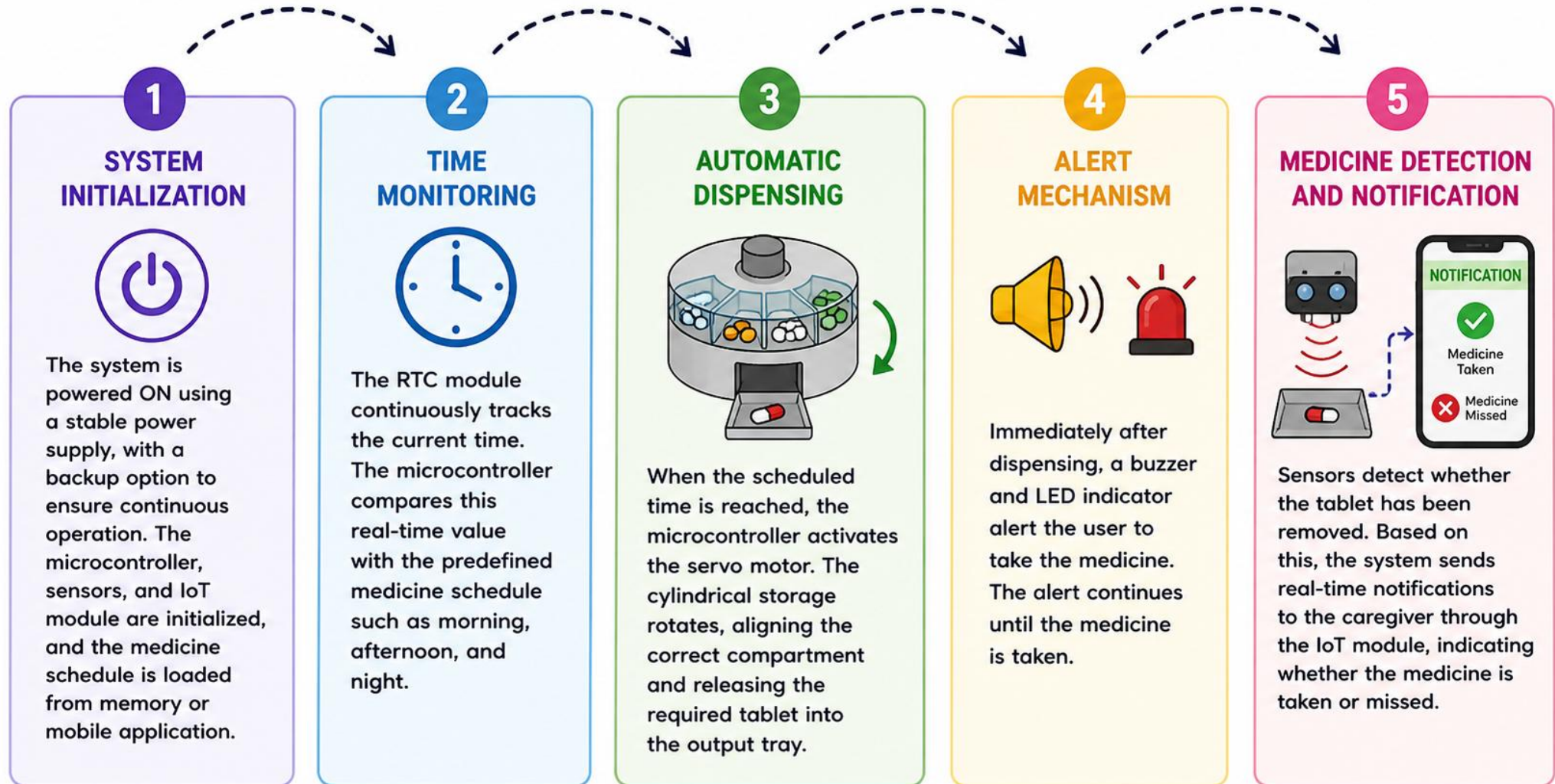


SUMMARY

These four modes ensure that the system works automatically, allows manual control, monitors medicine intake and alerts caregivers, and saves power for reliable and efficient operation.

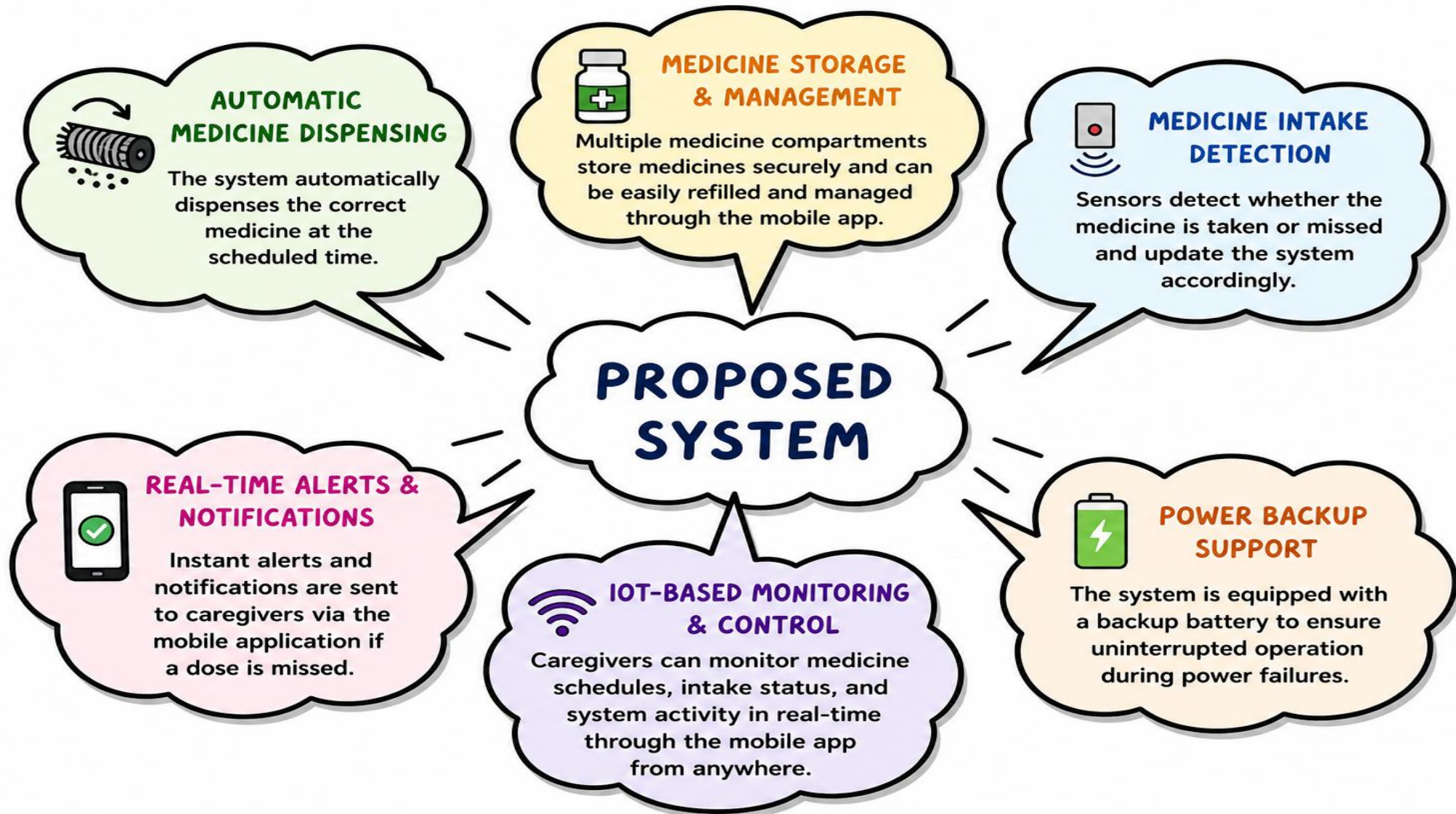
WORKING

WORKING OF THE PROJECT

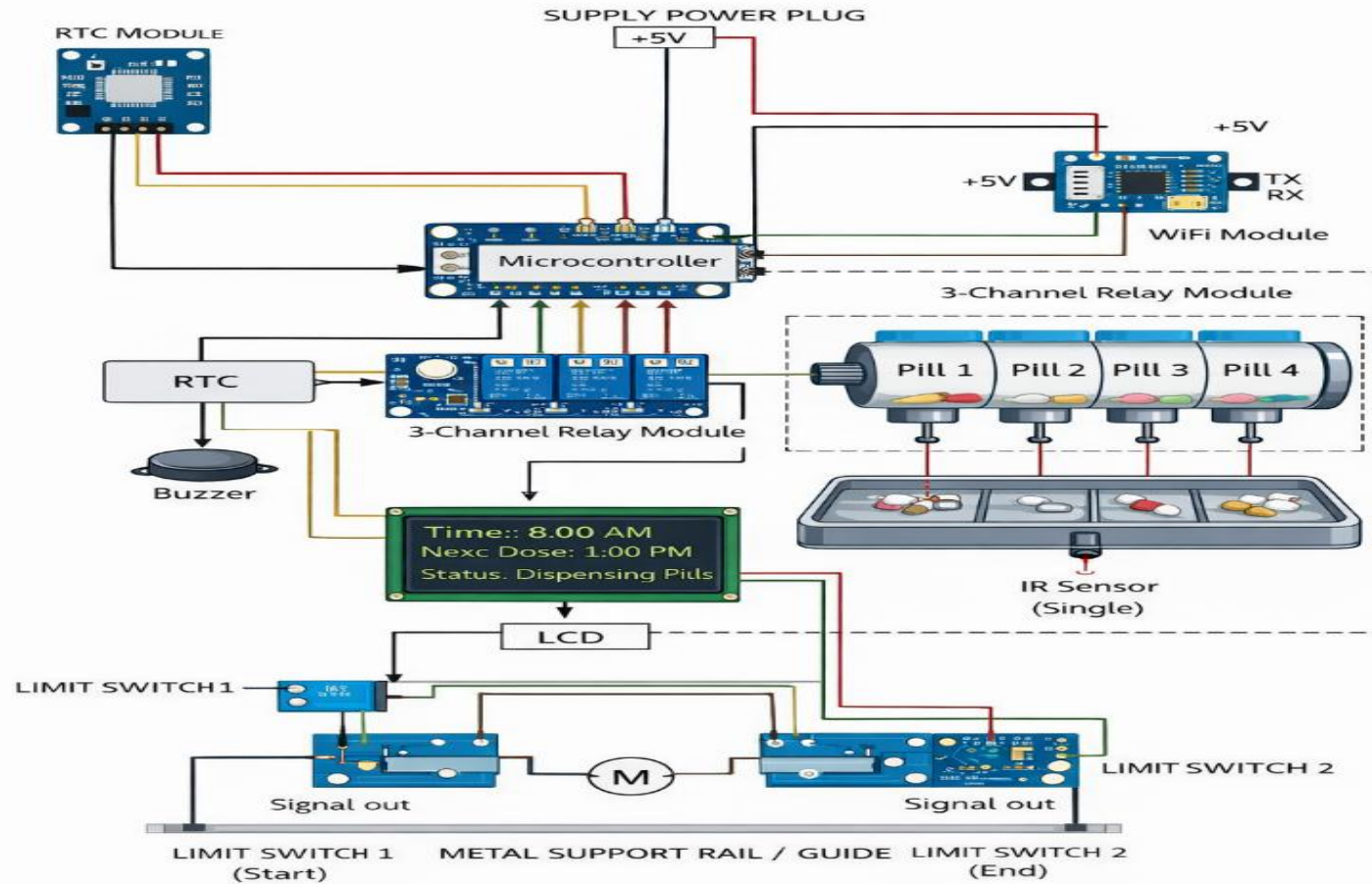




PROPOSED SYSTEM



CIRCUIT DIAGRAM OF SEWER NOVA



RESULTS AND DISCUSSION



Create New Alert

Date:

Time (Hr : Min):

 :

Select Tablet:

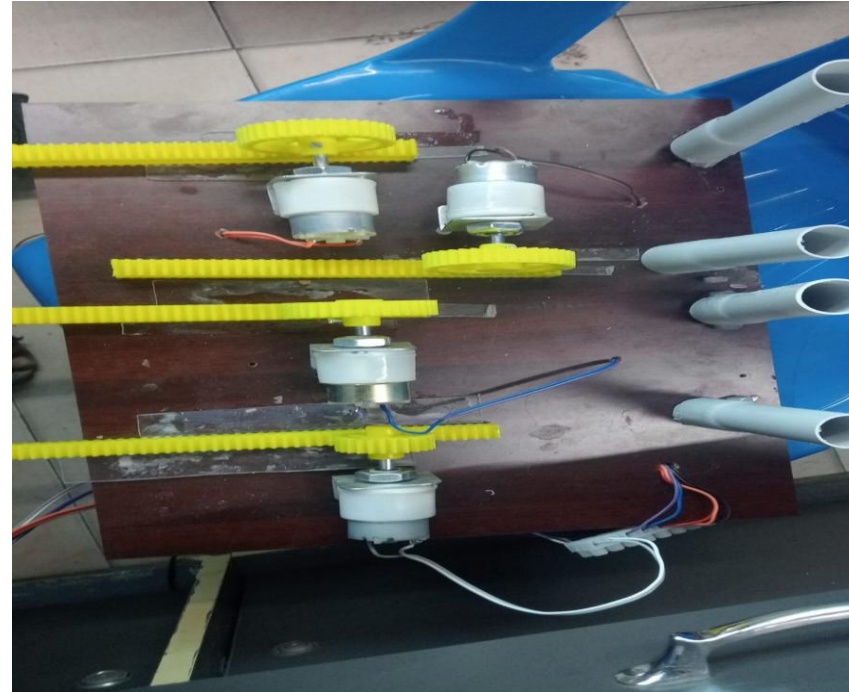
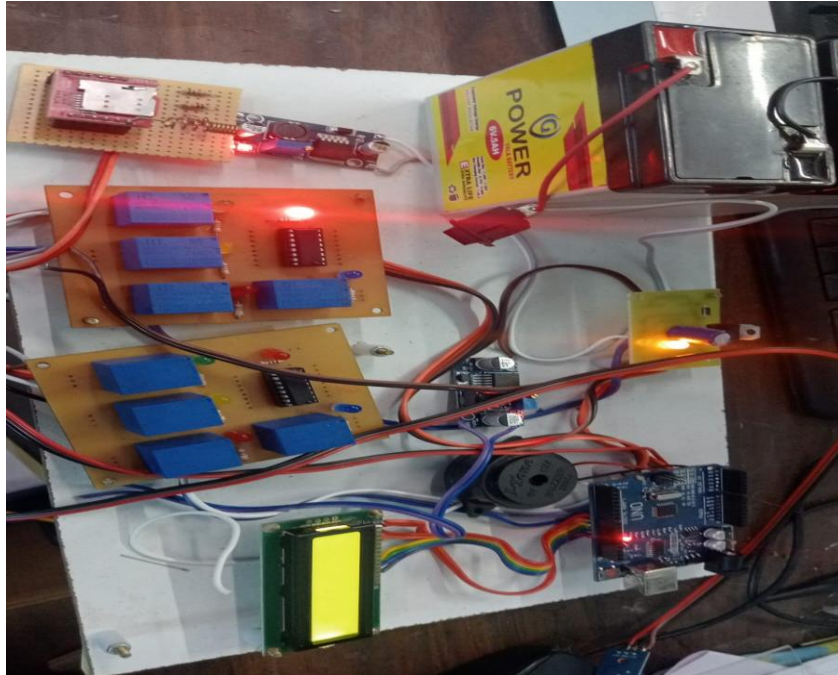
Description:

Add Alert

Print List

ID	Target	Date & Time	Description	Status	Action
1	Tablet 1	2026-04-03 18:05:00	test	MISSED	Delete
3	Tablet 1	2026-04-04 18:13:00	rr	COMPLETED	Delete
6	Tablet 1	2026-04-06 10:24:00	Tablet	MISSED	Delete
7	Tablet 1	2026-04-06 10:26:00	Tablet	MISSED	Delete

EXPERIMENTAL SETUP



INNOVATION AND UNIQUENESS



- Automated medicine dispensing system with cylindrical storage, ensuring accurate release of tablets at scheduled times without human intervention.
- Integrated sensor-based detection system to confirm whether the medicine is taken or missed, improving reliability and safety.
- Real-time IoT-based monitoring and notification system that updates caregivers instantly about medication status.
- Smart alert mechanism using buzzer and visual indicators to ensure users do not miss their medication.
- User-friendly mobile/web application interface for easy scheduling, control, and monitoring of medicine intake.

CONCLUSION



- To sum up, the Smart Automated Medicine Dispensing and Alert System is an effective, reliable, and economical solution for modern healthcare challenges.
- It integrates automation, sensing, and IoT communication technologies to ensure timely and accurate dispensing of medicines with minimal human intervention.
- The system not only improves medication adherence but also enhances patient safety by reducing the risk of missed doses and incorrect intake.
- With its user-friendly design, real-time monitoring, and alert features, it supports independent living for elderly and visually impaired individuals while reducing the burden on caregivers.
- Overall, this project represents a significant step toward smarter, safer, and more efficient healthcare management systems.

FUTURE SCOPE



- The IoT-based Smart Automated Medicine Dispensing and Alert System holds significant potential for future enhancements, making it a strong foundation for next-generation smart healthcare solutions.
- Future systems can be integrated with advanced sensors to monitor medicine levels, detect user interaction more accurately, and ensure precise dispensing of tablets.
- Additionally, incorporating features such as voice assistance, biometric identification, and health monitoring sensors (like heart rate and temperature) will improve usability, safety, and personalized care, enabling a more intelligent and comprehensive medication management system.

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OUTCOME



Our Project ,Title “Sewage Robot”, has been selected for College Seed fund of 5000, and it will be filed for a patent through our Institute IPM Cell.



M.Kumarasamy
College of Engineering

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Thalavapalayam, Karur - 639 113, TAMILNADU.



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